Nikita Kirnosov

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Skills

Languages C/C++, Python, SQL, R, Java, Bash

Techniques Pattern recognition, anomaly detection, clustering, classification, functional minimization, parallelization

Experience

2016 Apr – Present **Software Engineer**, *Google - YouTube*, San Bruno, CA, 40 hours/week.

Built storage abstraction microservices framework with no-downtime migration support (C++).
- Designed storage management framework allowing the user to create a full set of standard RESTful APIs for a defined resource in several hours regardless of the storage type.
- Performed video metadata migration between Vitess and Spanner with no downtime, data loss, or client latency increase.

Improved YouTube Public Data API performance, usability, and security (Python).
- Performed data access patterns analysis to identify convenience methods to be added to the API, improving performance and simplifying client side code.
- Built traffic analysis dashboards utilizing Fourier transform to visualize client requests in frequency domain and allow for simple in-service fraud detection.
- Enforced end user credential checking to prevent private data leaks.

2010 Aug – 2016 Jan **Teaching Assistant**, *University of Arizona*, Tucson,AZ, 20 hours/week.

Developed Fortran code modeling the molecular system total wave function as an optimal combination of explicitly correlated Gaussian functions. Generalized the approach to be applied to exotic atoms and molecules and extended it to excited states. The results are published in over a dozen peer-reviewed papers and presented on multiple physics and supercomputing conferences.
- Developed course curriculum and trained teaching assistants
- Taught advanced level courses
- Improved the code scalability and memory requirements by combining MPI and OpenMP frameworks
- Designed the stochastic basis function selection algorithm, which boosted the calculation accuracy by 3 orders of magnitude for electronic and 7 orders for muonic systems at the cost of insignificant number of additional operations
- Tuned the performance for changing computer architectures

2010 Jan–May **Intern**, *Joint Institute of Nuclear Research*, Dubna,Russia.

Contributed to muon detector for CMS software development by assessing signal processing code and vectorizing it

Side Projects

2015 August **WordSuggest**.

Performed n-gram tokenization of the HC Corpora corpus, built frequency tables and implemented simple back-off algorithm. To improve the performance for trending phrases, developed an algorithm to process Google search results page to suggest a next word in a phrase in real time.

2015 June **AskLocal**.

Developed a weather estimation app for any date during the next year for motorcycle around-the-world tour, which allowed to decrease equipment budget by 15%. The app retrieves a historical weather for a specified date and uses linear regression to extrapolate continuous parameters; random forest with 10-fold bootstrap is trained on each data set to predict precipitation events based on extrapolated parameters. Prediction results were on level of 7-day forecast accuracy.

2015 May **LecLab**.

Built an app to identify inconsistent grading. The grades data were retrieved from the course website and students' performance on computer-graded assignments was compared to performance on human-graded. Obtained insights lead yo lab performance assessment system reorganization, which allowed to reduce grades variability by 27%.

Education

2010–2015 **Ph.D.**, *University of Arizona*, Tucson, AZ – Physics.

2005–2010 **B.S. (Hons.)**, *Saratov State University*, Saratov, Russia – Physics.